

Cauchy-Euler equation

this is how a Cauchy-Euler equation looks like:

$$ax^2y'' + bx'y' + cy = 0$$

- notice that the power and the order of derivative on each term is the same this is how you identify a Cauchy-Euler equation

when you have two different solutions ex: $m_1 = 1$, $m_2 = 3$ use this eqn:

$$y = C_1 x^{m_1} + C_2 x^{m_2}$$

when you have two repeated roots ex: $m_1 = 1$, $m_2 = 2$ use this eqn:

$$y = C_1 x^{m_1} + C_2 x^{m_1} \ln(x)$$

when you have complex conjugates ex: $m_1 = \alpha \pm i\beta$ use this eqn:

$$y = x^\alpha [C_1 \cos(\beta \ln(x)) + C_2 \sin(\beta \ln(x))]$$

if the right hand side is zero all you would have to do is take the derivative of this $y = m^x$ and plug into equation and solve for m then write equation but if the right hand side is not equal to zero you still do the same thing but you must do a variation of parameters

$$x^2 y'' + 3xy' - 3y = 0$$

• notice this is equal to zero we will begin by taking the derivative of $y = x^m$:

$$y = x^m, \quad y' = m x^{m-1}, \quad y'' = m(m-1) x^{m-2}$$

now plug these into the equation

$$x^2 (m(m-1) x^{m-2}) + 3x (m x^{m-1}) - 3(x^m) = 0$$

$$x^2 (m(m-1) x^m \cdot x^{-2}) + 3x (m x^m \cdot x^{-1}) - 3(x^m) = 0$$

$$\cancel{x^2} \cdot \cancel{x^{-2}} (m(m-1) x^m) + 3\cancel{x} \cdot \cancel{x^{-1}} (m x^m) - 3(x^m) = 0$$

$$\cancel{x^m} (m^2 - m + 3m - 3)$$

$$(m^2 + 2m - 3)$$

$$(m+3)(m-1)$$

$$m = -3, \quad m = 1 \quad \bullet \text{this is the first case}$$

$$y = C_1 x^{-3} + C_2 x$$

$$x^2 y'' - 3xy' + 4y = 0$$

$$y = x^m, \quad y' = m x^{m-1}, \quad y'' = m(m-1) x^{m-2}$$

$$x^2 (m(m-1) x^{m-2}) - 3x (m x^{m-1}) + 4(x^m) = 0$$

$$x^2 (m(m-1) x^m \cdot x^{-2}) - 3x (m x^m \cdot x^{-1}) + 4(x^m) = 0$$

$$\cancel{x^2} \cdot \cancel{x^{-2}} (m(m-1) x^m) - 3\cancel{x} \cdot \cancel{x^{-1}} (m x^m) + 4(x^m) = 0$$

$$x^m (m^2 - m - 3m + 4)$$

$$x^m (m^2 - 4m + 4)$$

$$m = 2, \quad m = 2 \quad \bullet \text{this is a case \#2}$$

$$y = C_1 x^2 + C_2 x^2 \ln(x)$$

$$x^2 y'' - 3x y' + 7y = 0$$

$$y = x^m, \quad y' = m x^{m-1}, \quad y'' = m(m-1) x^{m-2}$$

$$x^2 (m(m-1) x^{m-2}) - 3x (m x^{m-1}) + 7(x^m) = 0$$

$$x^2 (m(m-1) x^m \cdot x^{-2}) - 3x (m x^m \cdot x^{-1}) + 7(x^m) = 0$$

$$\cancel{x^2} \cancel{x^{-2}} (m(m-1) x^m) - 3\cancel{x} \cancel{x^{-1}} (m x^m) + 7 x^m = 0$$

$$x^m (m^2 - m - 3m + 7) = 0$$

$$\cancel{x^m} (m^2 - 4m + 7) = 0$$

$$\begin{array}{l} a = 1 \\ b = -4 \\ c = 7 \end{array} \quad \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \rightarrow \quad \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(7)}}{2(1)} \quad m = 2 \pm i\sqrt{3}$$

• this is a case #3

$$y = x^2 \left[C_1 \cos(\sqrt{3} \ln(x)) + C_2 \sin(\sqrt{3} \ln(x)) \right]$$

$$x^2 y'' + x y' - y = x^2 e^{2x}$$

• this is a non-homogenous Cauchy-Euler equation its not equal to zero

• We will begin by taking the derivative

$$y = x^m, \quad y' = m x^{m-1}, \quad y'' = m(m-1) x^{m-2}$$

$$x^2 (m(m-1) x^{m-2}) + x (m x^{m-1}) - (x^m) = 0$$

$$x^2 (m(m-1) x^m \cdot x^{-2}) + x (m x^m \cdot x^{-1}) - (x^m) = 0$$

$$\cancel{x^2} \cancel{x^{-2}} (m(m-1) x^m) + \cancel{x} \cancel{x^{-1}} (m x^m) - (x^m) = 0$$

$$x^m (m^2 - m - m - 1)$$

$$\cancel{x^m} (m^2 - 2m - 1)$$

$$(m-1)(m+1)$$

$$m = 1, \quad m = -1$$

$$y_c = C_1 x^1 + C_2 x^{-1}$$

• now label y_1 and y_2

$$y_1 = x, \quad y_2 = x^{-1}$$

get the equation into standard form:

$$\frac{x^2 y'' + x y' - y = x^2 e^{2x}}{x^2} = y'' + \frac{1}{x} y' - \frac{1}{x^2} = e^{2x}$$

apply the wronskian on y_1 and y_2

$$w = \begin{vmatrix} x & x^{-1} \\ 1 & -x^{-2} \end{vmatrix} = (x)(-x^{-2}) - (x^{-1})(1) = -x^{-1} - x^{-1} = -2x^{-1}$$

plug into u_1 and u_2 and take the integral

$$u_1 = \int \frac{-y_2 f(x)}{w} dx$$

$$u_1 = \int \frac{\cancel{x^{-1}} \cdot e^{2x}}{\cancel{2x^{-1}}} dx$$

$$u_1 = \frac{1}{2} \int e^{2x} dx \quad \begin{array}{l} u = 2x \\ du = 2dx \\ \frac{du}{2} = dx \end{array}$$

$$u_1 = \frac{1}{4} e^{2x}$$

$$u_2 = \int \frac{y_1 f(x)}{w} dx$$

$$u_2 = \int \frac{x e^{2x}}{-2x^{-1}} dx$$

$$u_2 = -\frac{1}{2} \int x^2 e^{2x} dx$$

$$uv - \int v du \quad \text{liPet}$$

$$u = x^2$$

$$dv = e^{2x}$$

$$du = 2x dx$$

$$v = \frac{1}{2} e^{2x}$$

$$\left(\frac{1}{2} x^2 e^{2x} - \frac{1}{2} \int e^{2x} \cdot 2x dx \right)$$

$$\left(\frac{1}{2} x^2 e^{2x} - \int e^{2x} \cdot x dx \right)$$

$$uv - \int v du \quad \text{liPet}$$

$$u = x$$

$$dv = e^{2x}$$

$$du = dx$$

$$v = \frac{1}{2} e^{2x}$$

$$\left(\frac{1}{2} x e^{2x} - \frac{1}{2} \int e^{2x} dx \right)$$

$$\frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x}$$

$$\left(\frac{1}{2} x^2 e^{2x} - \int e^{2x} \cdot x dx \right)$$

$$\frac{1}{2} x^2 e^{2x} - \left(\frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} \right)$$

$$- \frac{1}{2} \left(\frac{1}{2} x^2 e^{2x} - \frac{1}{2} x e^{2x} + \frac{1}{4} e^{2x} \right)$$

$$u_2 = -\frac{1}{4} x^2 e^{2x} + \frac{1}{4} x e^{2x} - \frac{1}{8} e^{2x}$$

now apply this equation: $y_p = u_1 y_1 + u_2 y_2$

$$y_p = \left(\frac{1}{4} e^{2x} \right) x + \left(-\frac{1}{4} x^2 e^{2x} + \frac{1}{4} x e^{2x} - \frac{1}{8} e^{2x} \right) \frac{1}{x}$$

$$y_p = \cancel{\frac{x}{4} e^{2x}} - \cancel{\frac{1}{4} x e^{2x}} + \frac{1}{4} e^{2x} - \frac{1}{8} x^{-1} e^{2x}$$

$$y_p = \frac{1}{4} e^{2x} - \frac{1}{8} x^{-1} e^{2x}$$

now write the general equation $y = y_c + y_p$

$$y = C_1 x + C_2 x^{-1} + \frac{1}{4} e^{2x} + \frac{1}{8} x^{-1} e^{2x}$$

$$x^2 y'' - x y' + y = 4x$$

$$y = x^m, \quad y' = m x^{m-1}, \quad y'' = m(m-1) x^{m-2}$$

$$x^2 (m(m-1) x^{m-2}) - x (m x^{m-1}) + (x^m) = 0$$

$$x^2 (m(m-1) x^m \cdot x^{-2}) - x (m x^m \cdot x^{-1}) + (x^m) = 0$$

$$\cancel{x^2} \cdot \cancel{x^2} (m(m-1) x^m) - \cancel{x} \cdot \cancel{x^{-1}} (m x^m) + (x^m) = 0$$

$$\cancel{x^m} (m^2 - m - m + 1) = 0$$

$$(m^2 - 2m + 1)$$

$$(m-1)(m-1)$$

$$m=1, m=1$$

$$y = C_1 x^1 + x^1 \ln(x)$$

$$y_1 = x, \quad y_2 = x \ln x$$

$$\frac{x^2 y'' - x y' + y = 4x}{x^2}$$

→

$$y'' - \frac{1}{x} y' + \frac{1}{x^2} y = \frac{4}{x}$$

$$W = \begin{vmatrix} x & x \ln(x) \\ 1 & \ln(x) + 1 \end{vmatrix} = (x)(\ln(x) + 1) - (x \ln(x))$$

$$\Rightarrow \cancel{x \ln(x)} + x - \cancel{x \ln(x)} = x$$

$$\frac{x}{f} \frac{W(x)}{g}$$

$$f' \cdot g + f \cdot g'$$

$$(1)(\ln(x) + x \left(\frac{1}{x}\right))$$

$$\ln(x) + 1$$

$$u_1 = \int \frac{-y_2 f(x)}{W}$$

$$u_2 = \int \frac{y_1 f(x)}{W}$$

$$u_1 = \int \frac{-\cancel{x} \ln(x) \cdot \left(\frac{4}{\cancel{x}}\right)}{x}$$

$$u_2 = \int \frac{\cancel{x} \cdot \left(\frac{4}{\cancel{x}}\right)}{x}$$

$$u_2 = \int \frac{4}{x} dx$$

$$-4 \int \frac{1}{x} \ln(x) dx \quad \begin{array}{l} u = \ln(x) \\ du = \frac{1}{x} dx \\ x du = dx \end{array}$$

$$u_2 = 4 \ln(x)$$

$$-4 \int \frac{1}{\cancel{x}} u \cdot \cancel{x} du$$

$$u_1 = -4 \frac{(\ln(x))^2}{2}$$

$$u_1 = -2 (\ln(x))^2$$

$$y_p = u_1 y_1 + u_2 y_2$$

$$y_p = (-2 \ln(x))^2 (x) + (4 \ln(x)) (x \ln(x))$$

$$y_p = -2x (\ln(x))^2 + 4x (\ln(x))^2$$

$$y_p = 2x (\ln(x))^2$$

$$y = y_c + y_p$$

$$y = c_1 x' + x' \ln(x) + 2x (\ln(x))^2$$